

Multidisciplinary
Minor-10
Internet of Things

Preface

The Internet of Things in Engineering consists of sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet together. IoT has a huge impact on the engineering fields such as Smart cities, Smart health, Smart agriculture, Cyber-security, Preventive Maintenance Programs, Real-Time Construction Management Solution, Workers' Safety, Machine Governance and Field testing etc. In view of the same, the Multidisciplinary Minor (MDM) in Internet of Things (IoT) is offered by Electronics and Telecommunication Engineering for students of branches from Computer, IT, AIML, Mechanical and Civil Programs.

Objectives:

1. To familiarize the students with the fundamentals of hardware and software required for designing an IoT system.
2. To introduce the students with the technologies those contribute in building an IoT application.
3. To acquaint the students regarding building an IoT system.
4. To familiarize the students with the use cases of IoT in Industry.

Outcomes:

On the completion of the course, students will be able to

1. Develop embedded applications using IoT development boards, sensors and actuators.
2. Choose the appropriate protocol, cloud service and data handling methods for an IoT application.
3. Develop skills in identifying areas where IoT can be deployed for smart applications.
4. Implement real field problem by gained knowledge of Industrial applications with IoT capability.

CURRICULUM STRUCTURE Internet of Things

Course Code	Course Name	Credit Scheme				Teaching Scheme (Hours/Week)				Evaluation Scheme and Marks							
		L	P	T	T o t a l	L	P	T	T o t a l	FA		SA	TW	PR	OR	Total	
										FA1	FA2						
Semester III																	
BET23M D11	Principles of Internet of Things	2	-	-	2	2	-	-	2	10	10	30	-	-	-	50	
Semester IV																	
BET24M D11	Technologies Enabling IoT	2	-	-	2	2	-	-	2	10	10	30	-	-	-	50	
Semester V																	
BET25M D11	IoT System Design	3	-	-	3	3	-	-	3	20	20	60	-	-	-	100	
BET25M D12	IoT System Design Lab	-	1	-	1	-	2	-	2	-	-	-	50	-	-	50	
Semester VI																	
BET26M D11	Industrial IoT	2	-	-	2	2	-	-	2	10	10	30	-	-	-	50	
Semester VII																	
BET27M D11/BET 28MD11	Capstone Project	-	4	-	4	-	8	-	8	-	-	-	100	-	50	150	
Total		9	5	-	14	9	10	-	19	50	50	150	150	-	50	450	

L-Lecture, P-Practical, T-Tutorial, FA-Formative Assessment, SA-Summative Assessment, TW-Term Work, OR-Oral, PR-Practical

MDM: Internet of Things	B. Tech. :Comp, Comp RL, CSE, IT, Mech, Civil					Semester: III	
Course:	Principles of Internet of Things					Code:	BET23MD11
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50
Prior knowledge of: Basic Electronics Engineering							
Is essential							
Course Objectives: 1. To study fundamental concepts of IoT 2. To study roles of sensors, actuators and development boards in IoT 3. To correlate the connection of smart objects and IoT access technologies.							
Course Outcomes: After the completion of this course, the students will be able to: 1. Analyze the role of IoT in societal benefits. 2. Characterize the key attributes of sensors and actuators along with their significance in diverse IoT context. 3. Compare and contrast Arduino, Raspberry Pi, and ESP8266 development boards based on their specifications and GPIO pin structures. 4. Assess the impact of data acquisition and integration strategies on IoT system performance and scalability.							
Detailed Syllabus							

Unit	Description:	Duration [Hrs]
I	Introduction to IoT Introduction to Internet of Things (IoT), Functional Characteristics, Architecture, Recent Trends in the Adoption of IoT, Role of cloud in IoT, Main design principles and needed capabilities, standards considerations. M2M and IoT Technology .Societal Benefits of IoT:- Health Care, Machine to Machine (M2M), Smart Transportation, Smart Living, Smart Cities, Smart Grid	08
II	Sensing and Actuation Role of Sensors and Actuators in IoT, Sensors- Definition, characteristics, types and their working- Soil moisture sensor, DHT11, Ultrasonic sensor, PIR sensor, sound sensor, colour sensor, LDR, accelerometer, Gyroscope, magnetometer etc. , Actuator- Definition, characteristics, types and their working- LED, Relay, DC motor, LCD etc.	06
III	Development boards for IoT Introduction to Arduino, Raspberry Pi, NodeMCU, Beaglebone, Jetson Nano- Their versions, models, physical design, specifications, GPIO pin structure, OS requirement, capabilities, reading datasheet	08
IV	IoT Application Development Frame work for IoT Applications-Implementation of Device integration, Data acquisition and Integration, Device data storage on cloud/local server, Authentication, authorization of Devices	08
Total		30
Text Books: 1. Raj Kamal , “ Internet of Things: Architecture and Design”, McGraw Hill.2nd edition June 2022. 2. Misra, Sudip., Mukherjee, Anandarup., Roy, Arijit. “Introduction to IoT”. India: Cambridge University Press, 2021.		
Reference Books: 1. Pethuru Raj, Anupama C. Raman ,” The Internet of Things Enabling Technologies, Platforms, and Use Cases”, Taylor and Francis group. February 2017 2. Peter Waher, “Mastering Internet of Things: Design and create your own IoT applications using Raspberry Pi 3”, First Edition, Packt Publishing, 2018.		
e-sources: 1. https://nptel.ac.in/courses/117/102/117102060/		

MDM: Internet of Things		B. Tech. :Comp, Comp RL, CSE, IT, Mech, Civil				Semester: IV	
Course:		Technologies Enabling IoT				Code:	BET24MD11
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50
Prior knowledge of: Fundamentals of IoT							
Is essential							
Course Objectives: 1. To get an exposure to the different IoT enabling technologies. 2. To understand handling huge amount of data for its analysis. 3. To learn the contemporary aspects like cloud computing.							
Course Outcomes: After the completion of this course, the students will be able to: 1. Choose the hardware required for building an IoT system 2. Comprehend the concepts of WSN and choose the required communication protocol 3. Identify the required cloud service based on the requirement 4. Analyze the big data.							
Detailed Syllabus							
Unit	Description:						Duration [Hrs]
I	Embedded Systems Introduction to IoT Boards, IoT deployment for Raspberry Pi /Arduino/Equivalent platform – Interfacing ofSensors and actuators, Communication: Connecting microcontroller with mobile devices –communication through Bluetooth and wi-fi						07

II	Wireless Sensor Network What is communication? Wired and wireless communication, types of wired and wireless communication, What is WSN? WSN node, Anatomy of the network node, Architecture of WSN, Types of WSN WSN standards: IEEE 802.15.4, Low rate WPAN, Zigbee, BLE, LoRA. Protocol stack of WSNs	08
III	Cloud Computing Introduction to Cloud Computing (concept, architecture, working) Introduction to Cloud service models, Cloud Platforms: Google App Engine, Amazon Web Services, Microsoft Azure Cloud services, Windows Azure Platform Appliance. Distributed Computing: Need, Distributed computing vs. Cloud computing	07
IV	Big Data Analytics Introduction to Big data, Types of data, Characteristics of Big data, Data handling technologies, Flow of data, Data acquisition, Data storage, Introduction to Hadoop, Introduction to Data Analytics, Types of dataanalytics-Local Analytics, Cloud analytics and applications	08
Total		30
Text Books: 1. Pethuru Raj and Anupama C. Raman, "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", CRC Press, 2017 2. Olivier Hersent, David Boswarthick, and Omar Elloumi, —The Internet of Things: Key Applications and Protocols, Wiley Publications, 2012 3. Kazem Sohraby, Daniel Minoli and Taieb Znati, — Wireless Sensor Networks Technology, Protocols, and Applications—, John Wiley & Sons, 2010.		
Reference Books: 1. Hakima Chaouchi, — The Internet of Things Connecting Objects to the Web ISBN : 978-1- 84821-140-7, Wiley Publications, 2013 2. Holger Karl and Andreas Willig, —Protocols and Architectures for Wireless Sensor Networks, John Wiley & Sons, Ltd, 2007. 3. Daniel Minoli, —Building the Internet of Things with IPv6 and MIPv6: The Evolving World of M2M Communications, ISBN: 978-1-118-47347-4, Wiley Publications, 2013		
e-sources: 1. https://onlinecourses.nptel.ac.in/noc22_cs53/preview 2. https://www.udemy.com/course/internet-of-things-iot-fundamental-course-iot-101-level/?couponCode=IND21PM		

MDM: Internet of Things		B. Tech. :Comp, Comp RL, CSE, IT, Mech, Civil				Semester: V	
Course:		IoT System Design				Code:	BET25MD11
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
3	3	-	-	20	20	60	100
Prior knowledge of: Fundamental knowledge of Sensors, Actuators and IoT boards							
Is essential							
Course Objectives: 1. To explain the key requirements for implementing IoT with Arduino Uno and Raspberry Pi developmentboards. 2. To describe the Node-RED tool and python code for designing the IoT applications in Raspberry PI. 3. To understand IoT security issues and concerns to create awareness							
Course Outcomes: After the completion of this course, the students will be able to: 1. Use Arduino Uno for designing the IoT applications. 2. Illustrate use of Raspberry Pi for designing the IoT applications. 3. Develop an IoT system with Node-RED tool using Raspberry Pi 4. Develop the logic for Python Programming. 5. Understand IoT security systems							
Detailed Syllabus							
Unit	Description:						Duration [Hrs]
I	Implementing IoT with Arduino Introduction to Arduino Platforms, Arduino Uno architecture, IDE setup, importing Arduino boards in Arduino IDE tool, Installation of Arduino libraries, Basics of Embedded C Programming, Interfacing of Sensors and Actuators with Arduino Uno.						08

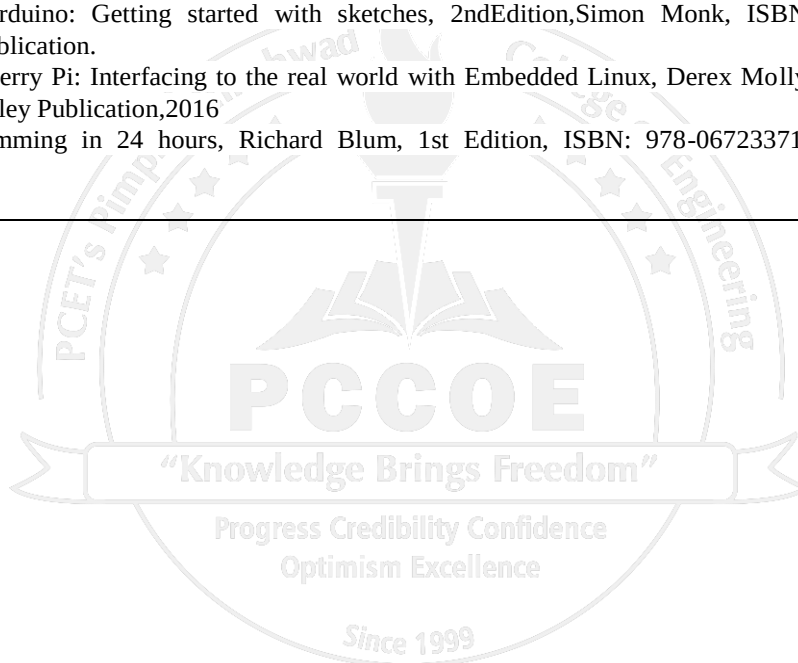
II	Implementing IoT with Raspberry Pi (RPI) Basic functionality of RPi board, RPi GPIO pins, Reading the datasheet of RPi setting up the board by installing OS, first boot and basic configuration of Rpi, Basic Linux Commands, Accessing RPi remotely using networking tools, Interfacing of Sensors and Actuators with RPi	08
III	Node-RED tool on Rpi Prerequisite for Node-RED, Installing and upgrading Node-RED, Running Node-RED applocally and as a service on network, auto-start on boot, opening the editor, installation of various libraries for Node-RED, Creation and deployment of flows, Case studies on debug window, HTTP server, chart, gauge, slider, dashboard form etc.	09
IV	Python Programming Introduction to Python programming language: Python Programming, Environment, Python Expressions, Strings, Functions, Data types in python, flow control, conditional statement, Loops, importing libraries	10
V	Securing the Internet of Things & Security Architecture Introduction, Security Requirements in IoT Architecture, Security in Enabling Technologies, Security Concerns in IoT Applications, Security Requirements in IoT, Insufficient Authentication/Authorization, Insecure Access Control, Threats to Access Control, Privacy, and Availability , Attacks Specific to IoT	10
Total		45
Text Books:		
1. Programming the Raspberry Pi: Getting Started with Python, Simon Monk, 3rd Edition, Tata McGraw Hill Publication. 2021 2. Pethuru Raj and Anupama C. Raman, "The Internet of Things: Enabling Technologies, Platforms, and Use Cases", CRC Press, 2017 3. Securing the Internet of Things, Shancang Li Li Da Xu, Syngress, 2017, Elsevier, ISBN: 978-0-12-804458-2		
Reference Books:		
4. Programming Arduino: Getting started with sketches, 2nd Edition, Simon Monk, ISBN: 978-1259641633, Tata McGraw Hill Publication. 5. Exploring Raspberry Pi: Interfacing to the real world with Embedded Linux, Derex Molly, 1st Edition, ISBN: 978-1119188681, Wiley Publication, 2016 6. Arduino Programming in 24 hours, Richard Blum, 1st Edition, ISBN: 978-0672337123, Sams Teach Yourself Publishing, 2014 7. Aditya Gupta, "The IoT Hacker's Handbook: A Practical Guide to Hacking the Internet of Things", ISBN: 1484242998, Apress publisher, 2019		
e-sources:		
1. https://nptel.ac.in/courses/108108179		

MDM: Internet of Things	B. Tech. :Comp, CompRL, CSE, IT, Mech, Civil					Semester: V	
Course:	IoT System Design Lab					Code:	BET25MD12
Credits	Teaching Scheme (Hrs. /Week)			Evaluation Scheme and Marks			
	Theory	Practical	Tutorial	TW	OR	PR	Total
1	-	2	-	50	-	-	50
Prior knowledge: 1. Programming languages 'C' & Python. 2. Interfacing with IoT boards Is essential							
Course Objectives: 1. To deliver practical knowledge of interfacing of sensors and actuators with Raspberry Pi. 2. To deliver practical hands-on using cloud-based interfacing with Arduino Uno, nodeMCU and RPi							
Course Outcomes: After learning the course, the students will be able to: 1. Select appropriate sensors, actuators, cloud platform for designing IoT applications. 2. Design IoT based Realtime monitoring system using an appropriate cloud platform. Automate the systems using various sensors, actuators and cloud platforms							
Detailed Syllabus							
Expt. No.	Suggested List of Experiments						
1	Basics of Internet of Things: Sensors, Actuators, IoT architecture and Gateway						
2	LED blinking with Arduino Uno.						
3	IoT sensors interface with Arduino (Temperature/Light sensors)						
4	Integration of Actuators with nodeMCU (Servo motor/Relay).						
5	Web-based home automation using Arduino Uno and Bluetooth						

6	Installation of Noobs/Raspbian OS in Raspberry Pi
7	Home Automation using Node-RED and Raspberry Pi
8	Temperature and Humidity monitoring using Node-Red and Raspberry Pi
9	Motion based intrusion detection system using nodeMCU/RPi
10	Automatic Irrigation System using nodeMCU/RPi
11	Installation of Smart Television setup using Raspberry Pi

References:

1. Daniel Minoli, "Building the Internet of Things with IPv6 and MIPv6: The Evolving World of M2M Communications", ISBN: 978-1-118-47347-4, Willy Publications, 2013
2. Programming Arduino: Getting started with sketches, 2nd Edition, Simon Monk, ISBN: 978-1259641633, Tata McGraw Hill Publication.
3. Exploring Raspberry Pi: Interfacing to the real world with Embedded Linux, Derex Molly, 1st Edition, ISBN: 978-1119188681, Wiley Publication, 2016
 Arduino Programming in 24 hours, Richard Blum, 1st Edition, ISBN: 978-0672337123, Sams Tech Yourself Publishing, 2014.



MDM: Internet of Things		B. Tech. :Comp, Comp RL, CSE, IT, Mech, Civil				Semester: VI	
Course:		Industrial IoT				Code:	BET26MD11
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	FA		SA	Total
				FA1	FA2		
2	2	-	-	10	10	30	50
Prior knowledge of: Fundamentals of IoT Is essential							
Course Objectives: 1. To provide knowledge for the design and analysis of Industry 4.0 Systems 2. To provide students with good depth of knowledge of Designing Industrial IoT Systems for various application							
Course Outcomes: After the completion of this course, the students will be able to: 1. Apply IoT principles to design Industrial 4.0 Systems for various applications. 2. Interface IIoT monitoring manufacturing processes by providing real-time insights and supporting data-driven decision-making. 3. Analyze next generation sensors and AR, VR concepts. 4. Implement real field problems of Industrial applications with IoT capability.							
Detailed Syllabus							
Unit	Description:						Duration [Hrs]
I	Introduction to Industrial IoT (IIoT) Systems: The Various Industrial Revolutions, Role of Internet of Things (IoT) & Industrial Internet of Things (IIoT) in Industry, Industry 4.0 revolutions, Support System for Industry 4.0						06
II	IIoT Data Monitoring:- IoT Gate way, IoT Edge Systems and It's Programming, Cloud computing, Real Time Dashboard for Data Monitoring, Data Analytics and Predictive Maintenance with IIoT technology						09

III	Cyber Physical Systems: Next Generation Sensors, Collaborative Platform and Product Lifecycle Management, Augmented Reality and Virtual Reality, Artificial Intelligence, Big Data and Advanced Analysis	08
IV	Industrial IoT- Applications:- Healthcare, Power Plants, Inventory Management & Quality Control, Plant Safety and Security (Including AR and VR safety applications), Facility Management	07
Total		30
Text Books: 1. Industry 4.0: The Industrial Internet of Things Alasdair Gilchrist Publications: (Apress Released June 2016) 2. “Industrial IoT: Technologies and Applications”; by Bhawani Shankar Chowdhry, Muhammad Shafique, and Habib M. Ammari (2nd Edition, 2021 Springer) 3. “Industrial IoT: Building Smarter Manufacturing and Energy Infrastructures”; by Krishnendu Chakrabarty and Elias Kyriakides (1st Edition, 2018 CRC Press)		
Reference Books: 1. Hakima Chaouchi, “ The Internet of Things Connecting Objects to the Web” ISBN : 978-1- 84821-140-7, Willy Publications. 2. Internet of Things- From Research and Innovation to Market Deployment; ; By Ovidiu Peter; River Publishers Series		
e-sources: https://nptel.ac.in/courses/108/105/108105088/ https://nptel.ac.in/courses/108/105/108105063/		

MDM: Internet of Things	B. Tech. :Comp, Comp RL, CSE, IT, Mech, Civil					Semester: VI	
Course:	Capstone Project/Seminar					Code:	BET27MD11/ BET27MD11
Credits	Teaching Scheme (Hrs./Week)			Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	Oral	TW		Total
4	-	8	-	50	100		100
Prior knowledge of: Basics of hardware, software and services required for an IoT application Is essential							
Course Objectives: 1. To test students' knowledge of course implementation. 2. To make students ready for development of an IoT system							
Course Outcomes: After the completion of this course, the students will be able to: 1. Apply Knowledge of IoT for solving real time issues. 2. Design a IoT system to solve Societal Issue. 3. Develop the project and communicate it to outside world.							
Detailed Guidelines							
1. The student should let the course instructor know in advance the intended topic of the project and seminar. 2. The students are encouraged to take projects for developing software solutions and hardware platforms using the concept of course taken under the certification. 3. The project guidelines mentioned in B.Tech (E&TC) structure will be followed for evaluation of performance and certification compliance. 4. The seminar guidelines mentioned in B.Tech (E&TC) structure will be followed for evaluation of performance and certification compliance.							
Detailed Syllabus:							

Sr. No.	Activity	Duration [Hrs]
1.	Week 1 & 2 : Mini-project guide allotment, finalization of topic and platform, Planning of the work	08
2.	Week 3&4: Literature review and specification and Methodology Finalization, Review 1 for finalization of topic and specification.	08
3.	Week 5&6 : Simulation of Idea on appropriate software tools and finalization of hardware platform	08
4.	Week 7 & 8 : understanding platform implementation and related software flow and execute block level design , Review 2 to understand the progress of the project	08
5.	Week 9 & 10: Mini Project Report writing and publication or copyright planning and execution.	08
6.	Week 11&12: Demonstration of Project work and Final Review for submission and term work compliances.	08
Total		48

